

1. Administrative & Study Identification

Full Study Title: A Phase III Multicenter Randomized, Double-Blind, Double-Dummy, Parallel-Group Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Teriflunomide in Adult Patients With Relapsing Multiple Sclerosis (RMS) **Short Title/Acronym:** FENhance 2 **Protocol Number:** GN42272 (2020-001168-28) **NCT Number:** NCT04586023 **Sponsor Name:** F. Hoffmann-La Roche Ltd (Roche) **Investigational Product:** Fenebrutinib (active comparator: Teriflunomide) **Phase of Development:** Phase III **Trial Status:** Active, Not Recruiting **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of fenebrutinib on the Annualized Relapse Rate (ARR) and disability progression in adult participants with Relapsing Multiple Sclerosis (RMS) over a minimum of 96 weeks. **Secondary Objectives:** To compare the effects of fenebrutinib versus teriflunomide on confirmed disability progression, brain lesion formation, and overall safety profile.

2.2 Background & Rationale Targeting Bruton’s Tyrosine Kinase (BTK) provides a novel, targeted mechanism of action in multiple sclerosis. Unlike non-specific B-cell depletion therapies (such as anti-CD20 monoclonal antibodies) which can result in lasting immune suppression and increased risk of infections, BTK inhibitors like fenebrutinib offer a highly targeted treatment approach aiming to reduce MS relapses and disability progression without causing broad immunosuppression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Teriflunomide) **Blinding:** Double-blind, Double-dummy (using dummy pills so neither doctors nor patients know the assigned group) **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 751 evaluable patients **Number of Sites:** Multicenter study running in 17 countries **Estimated Study Start:** Late 2020 **Estimated Primary Completion:** Fourth Quarter 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 55 years inclusive. 2. A diagnosis of RMS in accordance with the revised 2017 McDonald Criteria, with an Expanded Disability Status Scale (EDSS) score of 0 - 5.5 at screening. 3. Ability to complete the 9-hole Peg Test (9-HPT) for each hand in < 240 seconds and perform the Timed 25-foot Walk Test (T25FWT) in < 150 seconds. 4. For female participants of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive measures, and refrain from donating eggs. 5. For male participants: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive measures, and refrain from donating sperm.

4.2 Exclusion Criteria 1. Disease duration of > 10 years from the onset of symptoms and an EDSS score at screening < 2.0. 2. Female participants who are pregnant or breastfeeding, or intending to become pregnant. Male participants who intend to father a child during the study. 3. A diagnosis of primary progressive MS (PPMS) or non-active secondary progressive MS (SPMS). 4. Any known or suspected active infection at screening, including but not limited to a positive screening test for Hepatitis B and C, an active or latent or inadequately treated infection with tuberculosis (TB), or a confirmed or suspected progressive multifocal leukoencephalopathy (PML). 5. History of cancer including hematologic malignancy and solid tumors within 10 years of screening. 6. Known presence of other neurological disorders that could interfere with the diagnosis of MS or assessments of efficacy/safety, and clinically significant cardiovascular, psychiatric, pulmonary, renal, hepatic, endocrine, metabolic or gastrointestinal disease. 7. Rare hereditary problems of galactose intolerance, total lactase deficiency, or glucose-galactose malabsorption. 8. Hypoproteinemia. 9. Acute liver disease, chronic liver disease unless considered stable for > 6 months, presence of cirrhosis (Child-Pugh Class A, B, or C), or Gilbert's Syndrome. 10. Participants with significantly impaired bone marrow function or significant anemia, leukopenia, neutropenia or thrombocytopenia. 11. Any concomitant disease that may require chronic treatment with systemic corticosteroids or immunosuppressants during the course of the study. 12. History of alcohol or other drug abuse within 12 months prior to screening. 13. History of or currently active primary or secondary (non-drug-related) immunodeficiency, including known history of human immunodeficiency virus (HIV) infection. 14. Inability to complete an MRI scan. 15. Adrenocorticotrophic hormone or systemic corticosteroid therapy within 4 weeks prior to screening (inhaled and topical corticosteroids are allowed). 16. Receipt of a live-attenuated vaccine within 6 weeks prior to randomization. 17. Any previous treatment with immunomodulatory or immunosuppressive medication without an appropriate washout period.

4.3 Open-Label Extension (OLE) Inclusion Criteria 1. Completed the Double-Blind Treatment (DBT) phase of the study (remaining on study treatment; no other DMT administered) and who, in the opinion of the investigator, may benefit from treatment with fenebrutinib. 2. Participants randomized to the teriflunomide treatment arm during the DBT phase must undergo the accelerated teriflunomide elimination procedure (ATEP) prior to the first administration of open-label fenebrutinib. 3. For female and male participants: agreement to remain abstinent or use contraceptive measures, and refrain from donating eggs/sperm.

4.4 Open-Label Extension (OLE) Exclusion Criteria 1. Chronic liver disease unless considered stable for > 6 months. 2. Acute liver disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Daily oral administration (pills) of either fenebrutinib or teriflunomide, plus matching placebo dummies. **Route of Administration:** Oral **Duration of Treatment:** Minimum of 96 weeks (nearly 2 years) for the double-blind treatment (DBT) phase. Following the DBT phase (after disclosure of results), the Sponsor will determine whether to initiate the open-label extension (OLE) phase for participants to receive fenebrutinib.

5.2 Concomitant & Prohibited Therapy Permitted: Inhaled and topical corticosteroids. **Prohibited:** Adrenocorticotrophic hormone or systemic corticosteroid therapy within 4 weeks prior to screening. Receipt of a live-attenuated vaccine within 6 weeks prior to randomization. Any previous treatment with immunomodulatory or immunosuppressive medication without an appropriate washout period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -42 Days	Informed Consent, EDSS evaluation, MRI scan, Infection screening (HIV, Hep B/C, TB)
Baseline	Day 0	Randomization (1:1), First Dose
Interim	Weeks 2 to 24	Clinic visits every 2 weeks until Week 20, then at Week 24 (6 months) for safety and efficacy checks
Maintenance	Post Wk 24	Clinic visits every 3 months for the remainder of the 96-week double-blind period
End of Study	Week 96+	Final Double-Blind Visit, option to enter Open-Label Extension (OLE), or 2-month safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. **Annualized Relapse Rate (ARR)** over a minimum of 96 weeks. Measurement based on clinical evaluation of relapses by the investigator.

7.2 Secondary Endpoints 1. **Time to Onset of Composite 12-week Confirmed Disability Progression (cCDP12)** 2. **Time to Onset of Composite 24-week Confirmed Disability Progression (cCDP24)** 3. **Time to Onset of 12-week Confirmed Disability Progression (CDP12)**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via regular clinic visits (every 2 weeks initially, then every 3 months) and regular MRI scans to check for subclinical disease activity or neurological safety events. **Serious Adverse Events (SAEs):** Standard reporting timelines strictly applied (typically 24 hours), with specific monitoring for liver function and severe infections. **Data Monitoring Committee (DMC):** External independent committee reviewing safety data, particularly given the targeted BTK immunosuppressive mechanism.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints. **Sample Size Justification:** $N=751$ to achieve sufficient statistical power ($\alpha = 0.05$, $Power = 80\%$) to detect a significant difference in ARR and disability progression between fenebrutinib and teriflunomide. **Handling of Missing Data:** Standard multiple imputation or mixed-effect model repeated measures (MMRM) for continuous variable endpoints where appropriate.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring for early safety detection during the first 20 weeks, transitioning to quarterly monitoring. **Audit Trail:** Rigorous centralized MRI readings and eCRF capture to ensure blind maintenance (double-dummy) is intact.

11. Study Identifiers & Governance

Primary ID: GN42272 **Registry Number:** NCT04586023 **Sponsor/Lead Organization:** F. Hoffmann-La Roche Ltd (Roche) **Study Title:**

FENhance 2 **Official Regulatory Title:** A Phase III Multicenter Randomized, Double-Blind, Double-Dummy, Parallel-Group Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Teriflunomide in Adult Patients With Relapsing Multiple Sclerosis (RMS)

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) **Diagnosis Threshold:** 2017 Revised McDonald Criteria **Medical Methodology:** Targeted BTK Inhibition vs. Pyrimidine Synthesis Inhibition **Optimization Target:** Suppression of ARR and prevention of confirmed disability progression

13. Study Procedures & Methodology

Management Pathway: Double-blind active comparator administration requiring Accelerated Teriflunomide Elimination Procedure (ATEP) prior to any open-label fenebrutinib phase. **Evaluation Protocol (HCP):** Standardized EDSS scoring and functional mobility tracking (9-HPT, T25FWT). **Evaluation Protocol (Patient):** Routine MRI tracking to capture subclinical lesion burdens. **Quality Audit Mechanism:** Independent blinded EDSS raters and centralized MRI imaging reading core lab.

14. Observation & Follow-Up Schedule

Total Duration: Minimum 96 weeks (DBT) + Optional OLE phase **Onsite Visit Cadence:** Every 2 weeks up to week 20, then week 24, then every 3 months (Q12W) **Post-Treatment Follow-up:** 2 months post-cessation of all study treatments **Imaging Review Interval:** Pre-defined MRI schedules typically aligned with clinical quarterly/biannual milestones.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			____		[DD-MMM-YYYY]							